

0 Term Project

The term project consists of two parts:

- (a) A term paper of no more than four pages, following the style of *Physical Review Letters*. Figures count toward the page limit, but references do not. A shorter paper will not be viewed negatively.
- (b) A final presentation of approximately 12 minutes, following the style of an American Physical Society conference presentation, i.e. 10 minutes presentation + 2 minutes Q&A.

Because this course covers a broad range of topics and students have diverse academic backgrounds, you may choose any topic related to your field of study as long as it involves stochastic calculus. Possible projects include a Stochastic Differential Equation (SDE) model, a numerical simulation of a random phenomenon, a randomized approach to simulating a deterministic phenomenon, or a deterministic Partial Differential Equation (PDE) model for a Probability Density Function (PDF). Surprise me!

0.1 Term Project Proposal

In order to provide you some early feedback and guidance on the project, each student is asked to submit a proposal of no more than two pages. The proposal should describe the project's background, problem statement, proposed approach, expected results, and significance. Later, you may later expand the proposal into your final term paper. Later, this proposal can be expanded to the final term paper. I encourage you to think in terms of the Heilmeyer's Catechism (https://en.wikipedia.org/wiki/George_H._Heilmeyer), although some questions of the Heilmeyer's Catechism do not apply to the course project and can be ignored.

The deadline for the proposal is **11:59pm on October 16, 2026**. Please submit your proposal to Canvas.

0.2 Term Paper and Presentation

- (a) The final term paper is due on **11:59pm on December 11, 2026**. Please submit your term paper to Canvas.
- (b) The final presentation will be scheduled **during the last week of the semester**. The exact date and time will be announced later. Please submit your presentation slides to Canvas before the presentation.